Data Flippers

Linear Regression

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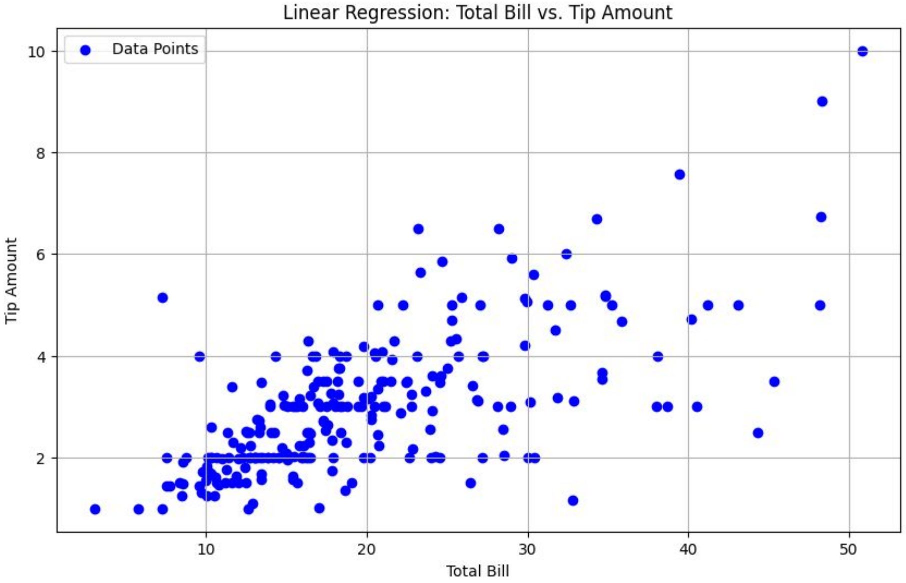
What is regression?

- Technique used for the modeling and analysis of numerical data
- Regression can be used for prediction, estimation, hypothesis testing, and modeling causal relationships.
- Exploits the relationship between two or more variables so that we can gain information about one of them through knowing values of the other



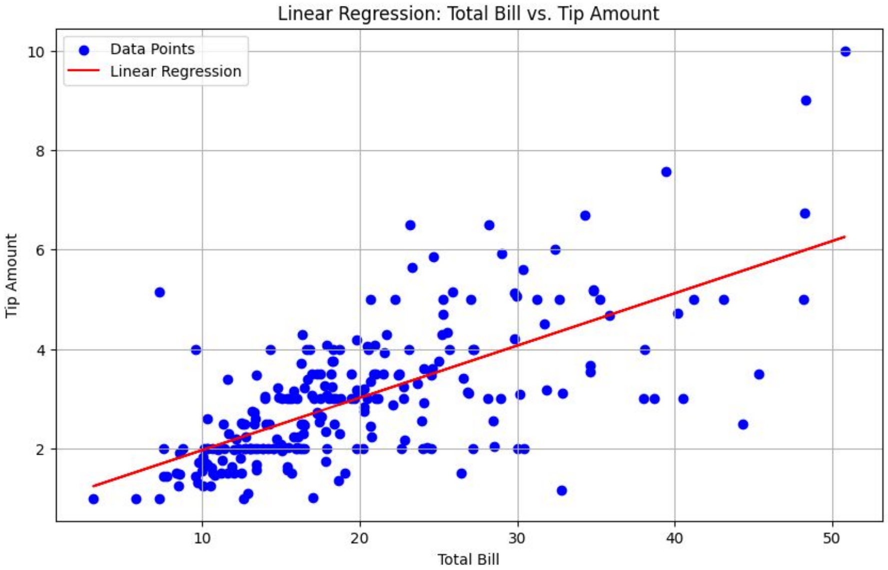
What is linear regression?

- Linear regression is the simplest form of machine learning. Some could even argue that linear regression isn't AI.

- Suppose we want to model the dependent variable Y in terms of three predictors, X1, X2, X3

$$\mathbf{Y = f(X1, X2, X3)}$$

- In short, $y = mx + b$



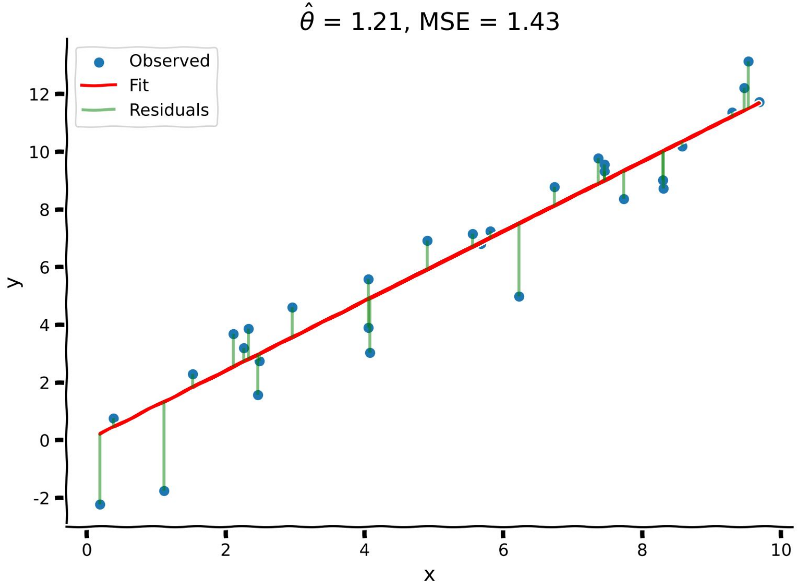
Talk to your neighbor

- When would linear regression be useful?
- Can you think of any limitations for linear regression?
- Can you think of any benefits for linear regression?

Measuring Accuracy :: MSE(Mean Square Error)

- Compute the raw distance from the line for each data point.

- Then take the mean(average) of the squared error of each point.



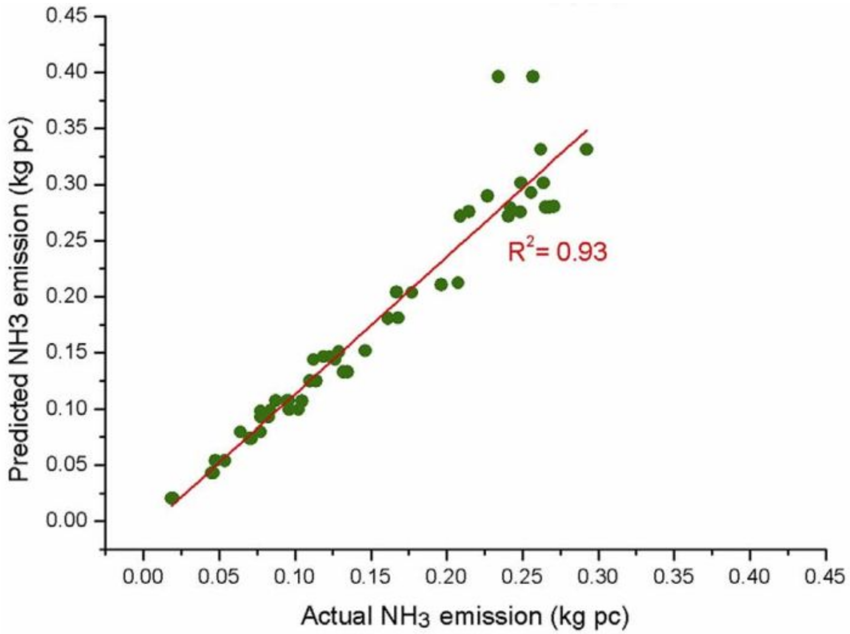
Measuring Accuracy :: R^2

- Shows how much the model "explains" the variability in the data.

- This Metric ranges from

0 (No correlation)

1 (Perfect correlation)



Measuring Accuracy :: Discussion

- Which is better? MSE or R^2 ? Or is both better?
- Can you think of a way to combine these two metrics?